

Matrix inequalities and norms

Sharp Schatten triangle constants for the arithmetic symmetric modulus

Difficulty: challenging **Importance:** interesting to specialist

Status: Partially resolved

Rating rationale: The unresolved Schatten parameters require sharp nonnormal extremizers; their direct impact is the quantitative theory of symmetric moduli.

Partial result — 2026-09-11

Partial result by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review: PASS for the stated partial result.**

For every $m \geq 2$, the exact dimension-two operator-norm constant is $c_{\infty}^{\text{sym}}(m, 2) = \sqrt{6\sqrt{3} - 9}$. This is also the sharp single constant valid simultaneously for every unitarily invariant norm on M_2 .

Still open: This does not determine each individual finite Schatten constant. The cases $1 < p < \infty$ remain open; the previously established trace endpoint and higher-dimensional operator endpoint are retained. The ratings apply to this surviving question.

Primary manuscript: [complete proof PDF](#), [standalone TeX](#), Theorem 1.1 and its proof; [authorship and scope](#). The [independent review](#) checks the full original argument and records its hash. The draft was AI-assisted; this is independent agent verification, not external human peer review or formal certification. [Submission record](#).

Problem statement

For $X \in \mathbb{C}^{n \times n}$ set $S(X) = ((X^*X)^{1/2} + (XX^*)^{1/2})/2$. Let $\|X\|_p = (\sum_i \sigma_i(X)^p)^{1/p}$ for $1 \leq p < \infty$, with $\|X\|_{\infty} = \sigma_1(X)$. Determine, for all integers $m, n \geq 2$ and all $p \in [1, \infty]$, the exact value

$$c_p^{\text{sym}}(m, n) = \sup_{(A_1, \dots, A_m) \neq (0, \dots, 0)} \frac{\|S(A_1 + \dots + A_m)\|_p}{\|S(A_1) + \dots + S(A_m)\|_p}, \quad A_j \in \mathbb{C}^{n \times n}.$$

The denominator is positive on the stated domain. A resolution should identify the value for the unsolved parameters while retaining known endpoint values.

Why it matters

This asks how much the symmetric positive representative of a matrix sum can exceed the sum of the individual representatives in Schatten norm. Sharp constants quantify the loss in this way of bounding nonnormal matrix sums.

References

1. T. Zhang, *Operator symmetric moduli and sharp triangle inequalities*, arXiv:2603.01046v1 (1 March 2026), Problem 1.17 and §8. [Primary text](#).
2. J.-C. Bourin and E.-Y. Lee, *Some hybrid matrix triangle inequalities*, arXiv:2606.29188v1 (28 June 2026), introduction and Theorem 1.3, for the dimension-independent majorization bound. [Primary text](#).

Status check — 2026-09-10

The source proves $c_1^{\text{sym}} = 1$ and $c_\infty^{\text{sym}} = \sqrt{2}$ when $n \geq 3$, and gives bounds for the other cases. Its latest version remains v1. Searches included *arithmetic symmetric modulus optimal Schatten constants*, *Zhang Problem 1.17 symmetric modulus*, and both paper titles with *2026 sharp*. No general determination was found. Problem 1.11's two-dimensional restriction is subsumed here and is not counted as an additional entry. Unlike the unitary-orbit conjecture, this problem asks for exact norm extrema, not a positive-semidefinite domination.

Audit update (2026-09-10): Rechecked Zhang Problem 1.17, including its endpoint values, and the June hybrid estimates. Searches for exact arithmetic-symmetric Schatten constants found no all-parameter answer; the displayed target includes the proved endpoints. This is a bounded literature check, not a proof that no solution exists.